

16 Mar. 2023

Spring 22-23 (102) First Mid

- $\lim_{x \rightarrow 0} (1 + 2x)^{\cot 3x}$
- $\int \frac{\cos^3 x}{\sin^3 x + 5 \sin x} dx$
- $\int_0^\pi e^{2x} \sin x \, dx$
- $\int \frac{dx}{x^2 \sqrt{9x^2 - 4}} dx$
- $\int (\sin 4x - \cos 3x)^2 dx$
- $\int_2^5 \frac{dx}{x^2 - 6x + 5}$

a) why is this Improper Integral?

b) Determine If the Integral convergent or divergent?

- $\lim_{x \rightarrow \infty} \frac{2x - \cos x}{x+1}$

a) Show that the limit is indeterminate of the form $\frac{\infty}{\infty}$

b) Why we can't use L'Hospital Rule to find the limit?